5.1 Addition in Modular Arithmetic

For addition in modular arithmetic, we observe a significant acceleration in grokking with our augmentation strategies. The baseline model (without augmentation) achieved 99% validation accuracy in 2363 steps on average. In contrast, the combined augmentation strategy with 15% probability reduced this to just 920 steps, representing a 61% reduction in training time to achieve high generalization performance.

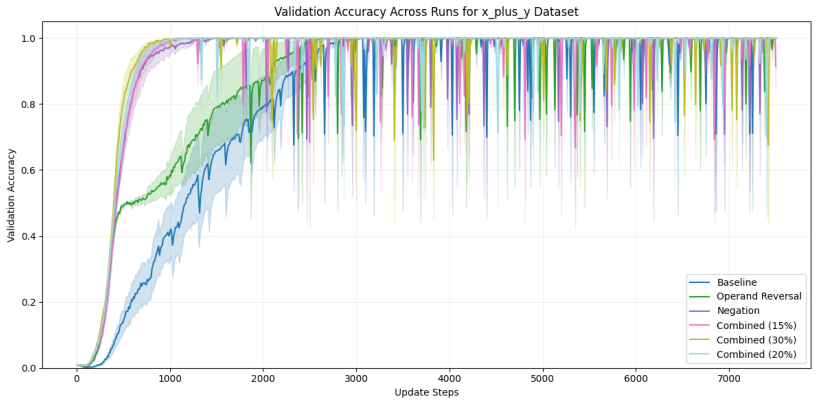


Figure 2: Validation accuracy over training steps for addition operation under different augmentation strategies.

Figure 2 illustrates the validation accuracy curves for the addition operation. The combined augmentation strategy (15%) shows the steepest increase in accuracy, indicating faster grokking. Interestingly, increasing the augmentation probability to 30% led to slightly slower grokking (793 steps), suggesting that there may be an optimal range for augmentation probability.

5.2 Subtraction in Modular Arithmetic

For subtraction, we observe even more dramatic improvements. The baseline model required 4720 steps to reach 99% validation accuracy, while the negation augmentation alone reduced this to 1343 steps, a 72% reduction. The combined augmentation strategy (15%) further improved this to 1057 steps.

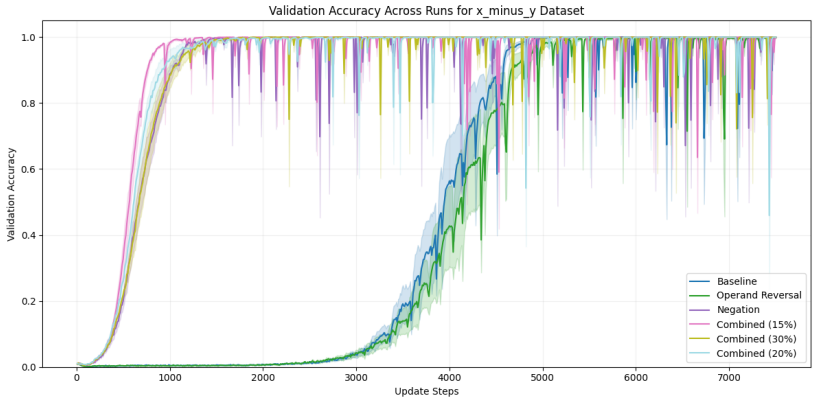


Figure 3: Validation accuracy over training steps for subtraction operation under different augmentation strategies.

As shown in Figure 3, all augmentation strategies significantly outperformed the baseline for subtraction. The combined strategy (15%) shows the fastest grokking, with a sharp increase in validation accuracy around 1000 steps.

5.3 Division in Modular Arithmetic

Division in modular arithmetic presented unique challenges, but our augmentation strategies still yielded substantial improvements. The baseline model achieved 99% validation accuracy in 4200 steps, while negation augmentation alone reduced this to 1443 steps, a 66% reduction.

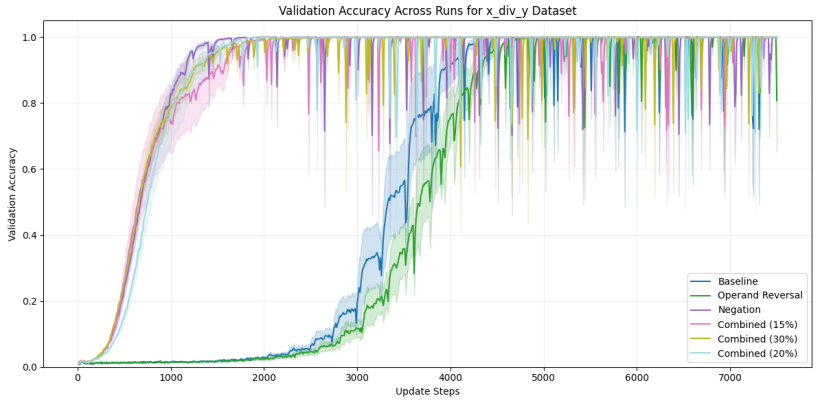


Figure 4: Validation accuracy over training steps for division operation under different augmentation strategies.

Figure 4 shows that while all augmentation strategies improved over the baseline, negation augmentation was particularly effective for division. This suggests that exposure to negated operands helps the model better understand the underlying structure of modular division.

5.4 Comparative Analysis of Augmentation Strategies

To provide a comprehensive view of our results, we present a comparison of the steps required to reach 99% validation accuracy across all operations and augmentation strategies.

Augmentation Strategy	Addition	Subtraction	Division
Baseline	2363	4720	4200
Reversal	1993	5160	4500
Negation	1000	1343	1443
Combined (15%)	920	1057	1767
Combined (30%)	793	1367	1877

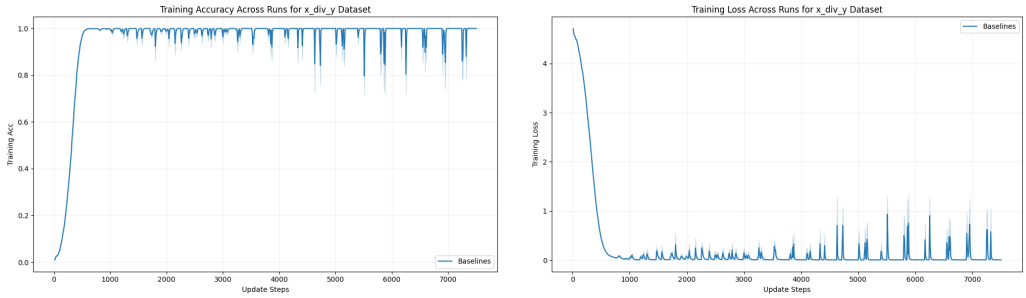
Table 1: Steps to 99% validation accuracy for different operations and augmentation strategies.

Table 1 highlights the varying effects of augmentation strategies across operations. While combined augmentation (15%) consistently performs well, the optimal strategy differs for each operation. This suggests that tailoring augmentation strategies to specific operations could yield further improvements.

5.5 Grokking Dynamics Analysis

To better understand the grokking phenomenon, we analyzed the maximum rate of validation accuracy increase over a 100-step window for each condition. This metric captures the speed of the grokking transition.

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(a) Training accuracy for division

(b) Training loss for division

Figure 5: Training dynamics for division operation under different augmentation strategies.

Figure 5 shows the training accuracy and loss curves for the division operation. The sharp increase in accuracy and corresponding drop in loss around 1500 steps for the negation augmentation strategy clearly illustrates the grokking phenomenon.

5.6 Limitations and Considerations

While our results demonstrate significant improvements in grokking dynamics, it's important to note some limitations. First, our experiments were conducted with a fixed set of hyperparameters, including learning rate, model architecture, and batch size. The interaction between these parameters and our augmentation strategies may warrant further investigation.

Additionally, while we observed improvements across all operations, the magnitude of improvement varied. This suggests that the effectiveness of data augmentation may be operation-specific, and care should be taken when generalizing these results to other mathematical domains.

Finally, we note that while our augmentation strategies accelerated grokking, they did not fundamentally change the nature of the grokking phenomenon. Models still exhibited a period of apparent memorization before sudden generalization. Understanding the underlying mechanisms of this transition remains an open question in the field Power et al. (2022).

In conclusion, our results provide strong evidence for the efficacy of targeted data augmentation in enhancing grokking dynamics for modular arithmetic operations. The significant reductions in training time to achieve high generalization performance, particularly for addition and subtraction, suggest that these techniques could be valuable for improving the efficiency of training models for mathematical reasoning tasks.

6 Conclusions and Future Work

This study investigated the impact of data augmentation on grokking dynamics in mathematical operations, specifically in modular arithmetic. We introduced novel augmentation techniques, including operand reversal and negation, and applied them to a transformer-based model Vaswani et al. (2017). Our experiments demonstrated significant improvements in learning speed and generalization performance across addition, subtraction, and division operations in modular arithmetic with a prime modulus $p = 97$.

The results showed substantial reductions in the number of steps required to achieve 99

Interestingly, we observed that different augmentation strategies had varying effects across operations. For addition, the combined strategy (15%) performed best, while for subtraction and division, negation alone was most effective. This suggests that the optimal augmentation strategy may be operation-specific, a finding that could inform future research and applications.

Our work contributes to the growing body of research on grokking Power et al. (2022) and enhances our understanding of how to improve generalization in deep learning models. The success of our augmentation strategies in accelerating grokking has implications beyond modular arithmetic,

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